

Task 24 – Data Quality Audit

Dataset: Sample - Superstore

1. Objective

The objective of this task was to audit the Superstore dataset using Python and Pandas, identify data-quality issues, document the findings in an issue log, and create a cleaned version of the dataset.

2. Tools Used

Python, Pandas, NumPy, and Google Colab.

3. Dataset Overview

Metric	Result
Rows	9,994
Columns	21
Missing values	0
Duplicate rows	0

4. Data Quality Checks

Check	Result
Missing values	0
Duplicate rows	0
Invalid Order Dates	0
Invalid Ship Dates	0
Ship Date before Order Date	0
Invalid Quantity	0
Invalid Discount	0
Invalid Sales	0
Postal Code format	449 records required standardization

5. Key Finding

The audit identified 449 Postal Code values that were not stored as five-character ZIP-code strings. These were treated as a formatting/standardization issue rather than automatically deleting the records.

6. Cleaning Performed

A copy of the original DataFrame was created. Postal Code was converted to text and standardized to five characters using zero-padding where needed. No rows were removed because the audit found no missing values, exact duplicates, invalid dates, invalid quantities, invalid discounts, or non-positive sales values.

7. Output Files

- Superstore_Cleaned.csv – cleaned Superstore dataset.
- Superstore_Issue_Log.csv – audit findings and issue counts.

8. Conclusion

The Superstore dataset was successfully audited using Pandas. Most quality checks returned no issues. The main identified issue was Postal Code formatting, which was standardized in the cleaned dataset. The audit results were documented in a separate issue log.